Textbook Template

Textbook Template

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Place a dedication here.

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Preface

This book template provides a starting point upon which authors may freely build to generate their own textbook entirely in LaTeX. We used this setup for Algorithms for Optimization, and have continued to refine it for a new textbook on decision making under uncertainty. The template allows for the direct compilation of a print-ready PDF, including support for figures, examples, and exercises.

This template is intended for textbook authors. Use of the template assumes prior expose to LaTeX. This template uses the `tufte-book` class to provide wide side margins.

Fundamental to our textbooks are the algorithms, which are all implemented in the Julia programming language. We have found the language to be ideal for specifying algorithms in human readable form. Algorithms can be typeset in algorithm blocks, tested, and used to generate figures. We use pygments and pythontex, including a custom pygments lexer and style.

We hope that you find this useful.

Mykel J. Kochenderfer
Tim A. Wheeler
Stanford, Calif.
June 20, 2019

Ancillary material is available on the template's webpage:
`https://github.com/sisl/textbook_template`

Acknowledgments

The authors wish to thank the many individuals who have provided valuable feedback on early drafts of our manuscript, including Adam Adamson, Barry Barryson, Cindy Cindyson, and Daryl Darylson. In addition, it has been a pleasure working with our editor from the MIT Press in preparing this manuscript for publication.

The style of this book was inspired by Edward Tufte. Among other stylistic elements, we adopted his wide margins and use of small multiples. In fact, the typesetting of this book is heavily based on the Tufte-LaTeX package by Kevin Godby, Bil Kleb, and Bill Wood.

We have also benefited from the various open source packages on which this textbook depends. The typesetting of the code is done with the help of pythontex, which is maintained by Geoffrey Poore. Plotting is handled by pgfplots, which is maintained by Christian Feuersänger. The book's color scheme was adapted from the Monokai theme by Jon Skinner of Sublime Text (`sublimetext.com`). For plots, we use the viridis colormap defined by Stéfan van der Walt and Nathaniel Smith.

PART I:

AWESOME CONTENT

1 Introduction

Welcome to the book template! This template is based on our previous book.[1]

[1] M. J. Kochenderfer and T. A. Wheeler, *Algorithms for Optimization*. MIT Press, 2019.

1.1 Using Julia

In this section we provide some examples of how Julia algorithms can be typeset, tested, etc. We begin with a Julia algorithm, algorithm 1.1. We use our custom `algorithm` environment with a pythontex `juliaverbatim` environment inside to typeset the algorithm. A caption provides some additional information, and shows up in the margin.

```
function sample_function(x, a)
    if x > a
        return log(x)
    else
        return x + log(a) - a
    end
end
```

Algorithm 1.1. A sample function that takes in an evaluation scalar `x` and a scalar parameter `a`.

We can add a test for our algorithm in the source code. This test does not show up when you compile. You can run all tests by executing `jl runtests.jl`.

The code can be executed when creating figures. Figure 1.1 is a standard inline figure that executes Julia code using `PGFPlots.jl` to produce a TEXfile in the `fig` directory. This file is then compiled into the PDF.

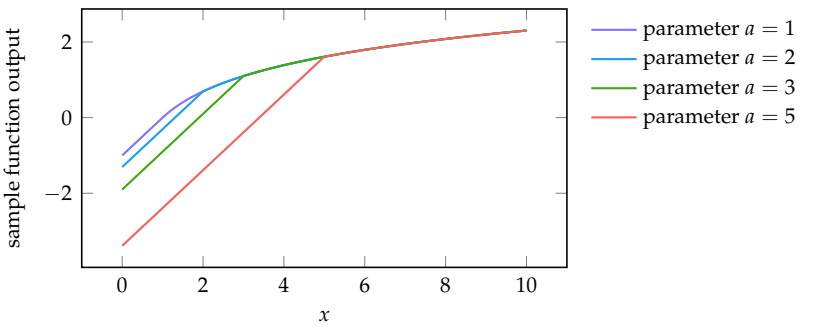


Figure 1.1. Curves obtained when using `sample_function` for several different values of `a`.

Of course, there is nothing stopping you from inserting tikzpictures directly, as shown in figure 1.2.

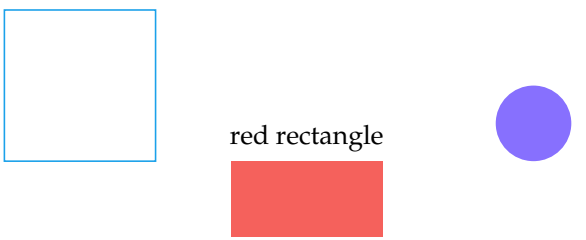


Figure 1.2. A figure made using a `tikzpicture` environment.

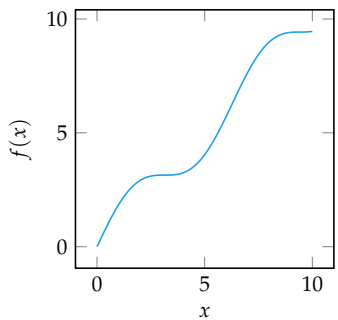


Figure 1.3. A `marginfigure` shows up in the margin.

Figures can also be placed in the margins, as with figure 1.4.

The `pythontex` package also supports the `juliaconsole` environment, as shown below. These environments act like the julia REPL, showing code as it is entered and executed.

1.2 *Custom Environments*

We created two custom environments in addition to the `algorithm` environment when writing *Algorithms for Optimization*. The `example` environment creates an isolated section in a light-gray box with its own caption.

An example of the `example` environment. Place anything you want here, including figures.

Example 1.1. An example `example`.

We also have `question` and `solution` environments, typically used at the end of each chapter. The solutions can automatically be placed at the end of the book by using `solutionappendix`.

Exercise 1.1. What is the meaning of life?

Solution: The meaning of life is 42.

1.3 Math Notation

Algorithms for Optimization uses lowercase characters with bold face for vectors, such as $\mathbf{x}$, and uppercase characters with bold face for matrices, such as $\mathbf{\Sigma}$. Scalars are lowercase with normal face, such as λ.

The contents of vectors and matrices can be typeset using `bmatrix`:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}^{-1} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \tag{1.1}$$

We often write out vectors horizontally in order to save space. In these cases, we use comma-separate entries:

$$\mathbf{x} = [x_1, x_2] \tag{1.2}$$

We construct an optimization problem as follows:

$$\begin{aligned} \underset{\mathbf{x}}{\text{minimize}} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & g(\mathbf{x}) \leq \mathbf{0} \\ & h(\mathbf{x}) = \mathbf{0} \end{aligned} \tag{1.3}$$

See `tufte_algorithm_book.cls` for some additional math utilities, such as $\mathcal{N}(\mu, \nu)$ and $|\mathcal{A}|$.

1.4 Code Replacement

Sometimes the function you want typeset is almost, but not quite, the function that you want to also execute or test. For example, you may want to inject an extra line to store data into a global variable, or you may wish to replace a particular function with a specialized version.

For such cases we have a simple replacement feature built into the `pull_julia_code` script. The script searches for `"REPLACE \(.*\) \(.*\)"` in the lines between the beginning of the `algorithm` environment and the beginning of the `juliaverbatim` environment. It will then match the string between the first parentheses and replace it with the string between the second parentheses when exporting the code. The code block will still be typeset with exactly what is inside the environment — it is only the exported code that is changed.

This feature was intended to be used sparingly.

```julia
function replacement_example(x)
    return x^2
end
```

Algorithm 1.2. A sample typeset algorithm whose exported code differs from what is typeset. Here the `x^2` will be replaced with `sin(x)`.

1.5 *Conclusion*

That is more or less everything you need in order to get started writing your own textbook with the `tufte_algorithm_book` style. We hope you find it useful.

If you have any questions or find improvements, please create issues on our GitHub repo: `https://github.com/sisl/textbook_template`.

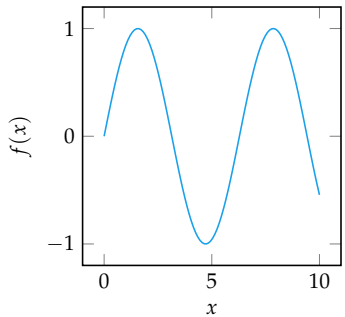


Figure 1.4. We find that when using algorithm 1.2, we actually get $f(x) = \sin(x)$ due to the replacement going on under the hood.

APPENDICES

A My Appendix Chapter

Oh yeah, and you can have chapters in the appendix.

References

1. M. J. Kochenderfer and T. A. Wheeler, *Algorithms for Optimization*. MIT Press, 2019 (cit. on p. 3).